

■ ResuBloom — Architecture, AI & Design Summary

Executive Overview: Full-Stack Architecture, Groq AI Integration & Key Engineering Decisions

1. System Architecture

ResuBloom is designed as a decoupled, high-performance **Full-Stack Client-Server Application**:

- **Frontend SPA (React 18 + Vite 6)**: Fast Single-Page Application with optimized Rollup vendor chunk splitting, responsive multi-step wizard, and real-time paper rendering canvas.
- **Backend REST API (Node.js + Express 5)**: Stateless microservice deployed on Render handling rate limiting, schema validation, and health diagnostics.
- **Database (MongoDB Atlas)**: User-isolated document database indexing resumes by clerkUserId and creation timestamp.
- **Authentication (Clerk)**: JWT-based zero-trust session validation securing backend endpoints.

2. AI Integration Strategy

The AI synthesis engine is built for **sub-2-second latency**, **zero hallucinations**, and **ATS compliance**:

- **Groq LPU™ & LLaMA 3.3 70B**: Leverages Groq's high-speed inference engine to generate and format full resumes in under 2 seconds.
- **Strict JSON Output Schema**: Uses native JSON mode (`response_format: { type: 'json_object' }`) with strict schema normalization in `groqService.js`, eliminating formatting parsing errors.
- **ATS Action-Verb Optimization**: Contextually converts passive phrases into high-impact executive verbs ('*Architected*', '*Spearheaded*', '*Accelerated*') while maintaining factual fidelity.
- **Quantifiable Impact Extraction**: Extracts and emphasizes metrics, scale, and performance outcomes to maximize keyword screening score.

3. Key Design Decisions & Strategic Rationale

Decision Area	Technical Choice	Strategic Rationale
Dual-State ATS Presentation	@media print CSS separation	Allows vibrant glassmorphism while editing, but strips all noise during PDF export for a 99% ATS-compliant white paper format.
Progressive UX Workflow	4-Step Wizard + Split Preview	Reduces cognitive load by segmenting input into Personal, Experience, Skills & Projects with instant visual paper feedback.
Botanical Floral Palette	Pastel Lavender & Soft Yellow	Replaces generic dark corporate styling with a calming, creative, and memorable brand identity (<i>ResuBloom</i>).
Zero-Trust Security	Clerk JWT Middleware	Authenticates at the Express middleware layer, guaranteeing strict privacy and isolation of candidate career documents.